



UTM
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MOBILE APPLICATION PROGRAMMING
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“TECHNICAL REPORT”
E-DOCS

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1. DEVELOPMENT APPROACH AND WORKFLOW

1.1 Methodology

The project followed an iterative and modular development approach, ensuring that each subsystem (authentication, document management, user management, UI) was built and validated independently.

1.2 Workflow Stages

1. Requirements Analysis

Identified core features: role-based access, document CRUD, user management, secure authentication.

2. Database & API Design

Designed PostgreSQL schema and REST API endpoints.

3. Backend Development (Node.js)

Implemented authentication, authorization, CRUD operations, and file handling.

4. Frontend Development (Flutter)

Built UI screens, navigation flow, and API integration.

5. Testing & Debugging

- Postman for API testing
- Flutter emulator + physical device testing
- Validation of file uploads, downloads, and JWT sessions

6. Refinement & Optimization

Improved UI responsiveness, error handling, and search/filter performance

2. SOFTWARE, TOOLS, AND FRAMEWORKS USED

2.1 Frontend (Flutter)

- Flutter SDK 3.10+
- Dart
- Packages:
 - http – REST API communication
 - provider – state management
 - file_picker – document upload
 - open_file – open downloaded files
 - path_provider – local storage

2.2 Backend (Node.js)

- Node.js v16+
- Express.js
- bcrypt – password hashing
- jsonwebtoken – JWT authentication
- multer – file uploads
- pg – PostgreSQL client

2.3 Database

- PostgreSQL 12+

2.4 Tools

- Postman
- Android Studio / VS Code
- Git & GitHub
- pgAdmin

3. DATABASE & DATA HANDLING DESIGN

3.1 Overview

The system uses a relational database with three primary entities:

Table	Description
Users	Stores user accounts, roles, and hashed passwords
Documents	Stores metadata for uploaded documents
Tags	Stores document categories (Procedure, Circular, Archive, Policy)

3.2 Key Design Principles

- Passwords stored using bcrypt hashing
- JWT tokens store user ID and role
- Documents stored in local filesystem, metadata in PostgreSQL
- Tags enforced using CHECK constraints

3.3 Example Schema Snippet

SQL

```
CREATE TABLE users (  
  id SERIAL PRIMARY KEY,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  role VARCHAR(10) CHECK (role IN ('admin', 'user')),  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE documents (  
  id SERIAL PRIMARY KEY,  
  title VARCHAR(255),  
  description TEXT,  
  filename VARCHAR(255),  
  tag VARCHAR(20) CHECK (tag IN ('Procedure', 'Circular', 'Archive', 'Policy')),  
  uploader_id INTEGER REFERENCES users(id),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

4. CRUD OPERATIONS EXPLAINED

4.1 Authentication

- Login (POST) /api/auth/login

4.2 Users (Admin Only)

- Create User – POST /api/users
- Read Users – GET /api/users
- Update User – PUT /api/users/:id
- Delete User – DELETE /api/users/:id

4.3 Documents

- Create Document – POST /api/documents
Uploads file + metadata using Multer.
- Read Documents – GET /api/documents
Supports search + tag filtering.
- Update Document – PUT /api/documents/:id
- Delete Document – DELETE /api/documents/:id
- Download Document – GET /api/documents/:id/download

5. IMPORTANT SOURCE CODE SNIPPETS

5.1 JWT Authentication (Node.js)

Javascript

```
const jwt = require('jsonwebtoken');

function generateToken(user) {
  return jwt.sign(
    { id: user.id, role: user.role },
    process.env.JWT_SECRET,
    { expiresIn: '24h' }
  );
}
```

5.2 Multer File Upload Configuration

Javascript

```
const storage = multer.diskStorage({
  destination: 'documents/',
  filename: (req, file, cb) => {
    cb(null, Date.now() + '-' + file.originalname);
  }
});

const upload = multer({ storage });
```

5.3 Flutter Login API Call

Javascript

```
const storage = multer.diskStorage({
  destination: 'documents/',
  filename: (req, file, cb) => {
    cb(null, Date.now() + '-' + file.originalname);
  }
});
const upload = multer({ storage });
```

5.4 Flutter Document Download

Javascript

```
final bytes = response.bodyBytes;
final file = File('${directory.path}/${filename}');
await file.writeAsBytes(bytes);
OpenFile.open(file.path);
```


6. TECHNICAL CHALLENGES & SOLUTIONS

6.1 Flutter Cannot Connect to Backend

Issue: Emulator cannot access localhost

Solution: Use 10.0.2.2 for emulator or PC IP for physical device.

6.2 File Uploads Errors

Issue: Missing directory or incorrect Multer config

Solution:

- Create /backend/documents
- Verify .env path
- Increase file size limit in Express

6.3 JWT Token Not Persisting

Issue: Token not stored properly

Solution: Use *SharedPreferences* and attach token to all API requests.

6.4 PostgreSQL Connection Errors

Issue: Wrong credentials or DB not running

Solution: Verify .env, ensure DB exists, test via pgAdmin.

6.5 Opening Downloaded Files on Android

Issue: Missing permissions

Solution: Add to AndroidManifest.xml:

Xml
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>